WELLSENSE · DESIGN DECISION RECORD

WellSense recommendations accepted

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DOCUMENT CONTROL

Document ID	WLS-DDR-002	Version	0.3
Status	Draft	Date	2026-10-02
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/wellsense		
Source	docs/decisions/0002-recommendations-accepted.md @ d54201d		

REVISION HISTORY

Version	Date	Author	Change
0.3	2026-10-02	Amish Chadha	O1 and O2 recorded as decided by Amish on 2026-10-02
0.2	2026-09-26	Amish Chadha	Budget set to $200 to cover the priced BOM: decided by Amish, 2026-09-26 (N1 closed)
0.1	2026-09-25	Amish Chadha	Record Amish's acceptance of all recommendations (2026-09-25), what changed in the repo, and the items still open

0002: Recommendations accepted

- **Date:** 2026-09-25
- **Status:** accepted. Every item below is "Decided by Amish, 2026-09-25: go with recommendation". Items without a recommendation were decided later: N1 (cost with the conduit) by Amish on 2026-09-26, and O1 and O2 by Amish on 2026-10-02 (WLS-DEC-001).

Context

On 2026-09-25 Amish wrote: "i accept all your recommendations, go with them across all repos." This record lists every WellSense item that carried a recommendation in `docs/REVIEW.md` (TRL 2 and TRL 3 sessions) and in WLS-DDR-001, and what changed in the repo because of it. Where a recommendation offered several options, the recommended option is the decision. Work that belongs to TRL 4 is recorded as decided but on hold, since TRL 4 is on hold by Amish's instruction.

Options considered

The options for each item are those listed in `docs/REVIEW.md` (sessions 2026-09-25, /populate and TRL 3) and in WLS-DDR-001.

Decision

Table 1. Items decided by Amish, 2026-09-25: go with recommendation.

#	Item	Decision	What changed in the repo
A1	D1, budget treatment of FieldNode (TRL 2 item 1)	Option (a): `budget_usd` covers WellSense parts only; FieldNode core costed in its own repo	Status wording in WLS-DDR-001, WLS-PRC-001 and REVIEW.md; no design change
A2	D2, sensor type (TRL 2 item 2)	Vented 4 to 20 mA transducer; absolute RS-485 sensor kept as a variant	Status wording only; the TRL 3 detail choices (24 V boost, 0.25 % class, two-point calibration) stand
A3	D3, range (TRL 2 item 3)	0 to 10 m, 0 to 20 m variant	Status wording only
A4	D4, access tube (TRL 2 item 4)	Own tube wherever a pump shares the casing	Status wording only
A5	D5, default interval (TRL 2 item 5)	15 min, 1 min in pumping tests	Status wording only
A6	D6, manual tape check (TRL 2 item 6)	Quarterly tape check with one shared tape	Status wording only
A7	D7, data ownership (TRL 2 item 7)	Levels only, community controls sharing, CSV export	Status wording only
A8	D8, FieldNode as the core	Reuse FieldNode for enclosure, power, radio and storage	Status wording only
A9	Cost at the design depth (TRL 3 item 3)	Option (b): raise `budget_usd` to $190 for a 30 m design well, $1.80 per metre beyond	`project.yaml` `budget_usd` $180 to $190; R12 restated in WLS-REQ-001 v0.4; README, precis and CAL updated
A10	Access tube bore (TRL 3 item 4)	Option (a): keep the 1 in tube and require a probe of 22 mm or less	`cad/src/model.py` probe 24 mm to 22 mm (radial clearance 1.3 mm to 2.3 mm); BOM line 1 respecified; R10 restated; STEP, STL, WLS-DWG-001 (Rev P2) and media regenerated
A11	Surface cable protection (TRL 3 item 5)	Option (a): galvanized conduit from the seal plate to the junction box, about $10 (estimate)	BOM line 14 added ($10.00); conduit modeled (`conduit` part); drawing note and callout; parts cost $187.60 to $197.60
A12	Hot-site rating (TRL 3 item 6)	Option (a): keep R15 at 55 °C and require FieldNode's sun shield at hot sites	R15 text in WLS-REQ-001 v0.4; precis; complete hot-site cost $331.60 with FieldNode's $8.00 shield (FND-DDR-002). FieldNode itself not edited
A13	Pumping-test uplinks (TRL 3 item 7)	On a public network send every 30 min during a test; use a TwinKit gateway where possible	R6 text; CAL-001 v0.2 [E4]: 30-reading batch every 30 min, 22.7 s a day at SF9, within 30 s fair use. Firmware is TRL 4 work and is on hold

No rewording of the pitch or problem line was recommended, so both are unchanged.

Table 2. Items open on 2026-09-25 (no recommendation); O1 and O2 decided by Amish on 2026-10-02 (WLS-DEC-001).

#	Item	Status
O1	First partner and region for co-design (WLS-DDR-001, O1). No preference stated.	Decided 2026-10-02: ACWADAM in Pune, India, as the first candidate to approach
O2	Battery-only FieldNode variant (WLS-DDR-001, O2). No recommendation to adopt.	Decided 2026-10-02: keep the standard core with its panel
N1	New: cost with the conduit. A9 and A11 together give $197.60 at 30 m against $190. Options: (a) raise `budget_usd` to $200; (b) keep $190 and state R12 at a 25 m design depth; (c) keep $190 and fit the conduit only where livestock or tampering is likely.	Decided by Amish, 2026-09-26: budget set to $200 (see below)

Budget approved, 2026-09-26

On 2026-09-26 Amish wrote: "i approve all the budget items."

- Budget set to $200 to cover the priced BOM: decided by Amish, 2026-09-26. This closes N1. `project.yaml` `budget_usd` 190 to 200; WLS-REQ-001 R12 target $200, status Not met to Met (budget holds to a 31.3 m probe depth, was 25.8 m); WLS-CAL-001 v0.3 (`sizing.py` and `results.csv` rerun); WLS-PRB-001, WLS-PRC-001, README, `bom/bom-notes.md` and the blueprint key figures updated. The BOM and geometry are unchanged.

Consequences

- `project.yaml` : `budget_usd` $190. `trl: 3` and `trl_target: 3` unchanged.
- Documents revised: WLS-PRB-001 v0.4, WLS-PRC-001 v0.4, WLS-REQ-001 v0.4, WLS-CAL-001 v0.2, WLS-DDR-001 v0.2; drawing WLS-DWG-001 Rev P2.
- Requirement status: 2 not met (R9 water safety, R12 cost at $197.60 against $190), 4 at risk (R4, R10, R15, R16), 3 not verifiable at TRL 3 (R5, R11, R13), 7 met.
- After the 2026-09-26 budget approval (WLS-CAL-001 v0.3): 1 not met (R9), 4 at risk, 3 not verifiable at TRL 3, 8 met.
- Cross-repo action (FieldNode, not edited here): confirm the shielded core's behaviour at 55 °C ambient, since WellSense R15 asks for 55 °C and FieldNode is rated to 45 °C ambient.
- TRL 4 remains on hold by Amish's instruction. The firmware uplink rule, the interface board build, supplier certificates and any purchasing are decided in principle but not started.